

MAT 310, Handout 21: Covariance and Linear Combinations of RVs (5.7, 5.8)

Question: We know how to determine if two random variables are independent, but how do we measure their dependence if they're not independent?

Covariance: Let $\mu_1 = E(Y_1)$ and $\mu_2 = E(Y_2)$. The covariance of Y_1 and Y_2 is given by

$$\text{Cov}(Y_1, Y_2) =$$

- If $\text{Cov}(Y_1, Y_2) > 0$, then as the value of Y_1 increases, in general the value of Y_2 also increases.
- If $\text{Cov}(Y_1, Y_2) < 0$, then as the value of Y_1 increases, in general the value of Y_2 decreases.

Correlation Coefficient: Let $\sigma_1 = \sqrt{V(Y_1)}$ and $\sigma_2 = \sqrt{V(Y_2)}$. The correlation coefficient of Y_1 and Y_2 is given by

$$\rho =$$

- $-1 \leq \rho \leq 1$
- $\rho > 0$ means that Y_1 and Y_2 are _____.
- $\rho < 0$ means that Y_1 and Y_2 are _____.
- $\rho = 0$ means that Y_1 and Y_2 are _____.

Theorem.

$$\text{Cov}(Y_1, Y_2) = E(Y_1 Y_2) - E(Y_1)E(Y_2)$$

Proof:

Theorem. If Y_1 and Y_2 are independent, then $\text{Cov}(Y_1, Y_2) = 0$.

Proof:

Note: The converse of this theorem is not necessarily true. That is, $Cov(Y_1, Y_2) = 0$ does NOT necessarily imply that Y_1 and Y_2 are independent.

Example 1. Find $Cov(Y_1, Y_2)$ given the following joint density function:

$$f(y_1, y_2) = \begin{cases} 2y_1, & 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1 \\ 0, & o.w. \end{cases}.$$

Example 2. Find $Cov(Y_1, Y_2)$ given the following joint density function:

$$f(y_1, y_2) = \begin{cases} 6(1 - y_2), & 0 \leq y_1 \leq y_2 \leq 1 \\ 0, & o.w. \end{cases}.$$

- $Cov(X, Y) = Cov(Y, X)$
- $Cov(Y, Y) =$
- $Cov(aX, bY) =$

- For any 2 RVs X and Y ,

$$V(aX + bY) =$$

$$V(aX - bY) =$$

- If X and Y are independent RVs, these expressions simplify as follows:

$$V(aX + bY) =$$

$$V(aX - bY) =$$

Let $U = \sum_{i=1}^n a_i X_i$ and $V = \sum_{j=1}^m b_j Y_j$. Then,

- $E(U) =$
- $V(U) =$
- $Cov(U, V) =$

Example 3. Suppose X_1 , X_2 , and X_3 are RVs with the following properties:

$$\begin{array}{lll} E(X_1) = -1 & V(X_1) = 1 & Cov(X_1, X_2) = -1 \\ E(X_2) = 0 & V(X_2) = 4 & Cov(X_1, X_3) = -2 \\ E(X_3) = 2 & V(X_3) = 9 & Cov(X_2, X_3) = 1 \end{array} .$$

Also, let $Y = 3X_1 + 2X_2 - X_3$ and $W = 2X_1 - 3X_2$. Find:

(a) $E(Y)$

(b) $V(Y)$

(c) $Cov(Y, W)$